There is a completely separable mad family of vector spaces

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Almost disjoint subspaces

Let $\mathbb{F}$ be a countable field (possibly finite). Let E be a $\mathbb{F}$-vector space with a Hamel basis $(e_n)_{n<\omega}$.

Definition

Let $V, W \subseteq E$ be two infinite-dimensional subspaces. We say that V, W are *almost disjoint* if $V \cap W$ is a finite-dimensional subspace of E.

For the rest of this talk, the zero vector will be ignored.

Definition

Let $\mathcal{A}$ be a family of infinite-dimensional subspaces of E. We say that $\mathcal{A}$ is *almost disjoint* if all subspaces in $\mathcal{A}$ are pairwise almost disjoint. We say that $\mathcal{A}$ is *maximal almost disjoint* (or just *mad*) if $\mathcal{A}$ is not strictly contained in another almost disjoint family of infinite-dimensional subspaces.

Definition

We define the cardinal invariant:

$$\mathfrak{a}_{\mathrm{vec},\mathbb{F}} := \min\{|\mathcal{A}| : \mathcal{A} \text{ is an infinite mad family of block subspaces}\}.$$

Lemma (Smythe, 2019)

$\aleph_1 \leq \mathfrak{a}_{\mathrm{vec},\mathbb{F}}$.

Block subspaces

Recall that E has a fixed Hamel basis $(e_n)_{n<\omega}$. Given a vector $x \in E$, we may write

$$x = \sum_{n<\omega} \lambda_n(x) e_n,$$

where only finitely many λ_n's are non-zero. We may then write:

$$\mathrm{supp}(x) := \{n < \omega : \lambda_n(x) \neq 0\}.$$

Notation

Given two vectors x, y we write:

$$x < y \iff \max(\mathrm{supp}(x)) < \min(\mathrm{supp}(y)).$$

Definition

An infinite-dimensional subspace $V \subseteq W$ is a *block subspace* if it has a *block basis*. That is, V is spanned by a *block sequence* $(x_n)_{n<\omega}$, i.e.:

$$x_0 < x_1 < x_2 < \cdots .$$

Note that the block basis is unique up to scalar multiplication, so we may conflate block subspaces with block sequences.

Fact

Every infinite-dimensional subspace of E contains an infinite-dimensional block subspace.

Consequently, if $\mathcal{A}$ is an almost disjoint family such that there is no block subspace that is almost disjoint with every element of $\mathcal{A}$, then $\mathcal{A}$ is mad.

Notation

Let $E^{[\infty]}$ be the set of infinite block sequences of E.

Notation

If $A = (x_n)_{n<\omega} \in E^{[\infty]}$, we write:

$$\langle A \rangle := \langle x_n : n < \omega \rangle = \operatorname{span}\{x_n : n < \omega\}.$$

Completely separable mad families (on ω)

Let $\mathcal{A}$ be an almost disjoint family on ω. The collection:

$$\mathcal{I}(\mathcal{A}) := \left\{ X \subseteq \omega : X \subseteq^* \bigcup \mathcal{F} \text{ for some finite } \mathcal{F} \subseteq \mathcal{A} \right\}.$$

is an ideal. We also let:

$$\mathcal{I}^+(\mathcal{A}) := [\omega]^\omega \setminus \mathcal{I}(\mathcal{A}),$$
$$\mathcal{I}^{++}(\mathcal{A}) := \{ X \subseteq \omega : X \cap A \text{ is infinite for infinitely many } A \in \mathcal{A} \}.$$

Note that $\mathcal{I}^{++}(\mathcal{A}) \subseteq \mathcal{I}^+(\mathcal{A})$, and $\mathcal{I}^{++}(\mathcal{A}) = \mathcal{I}^+(\mathcal{A})$ if $\mathcal{A}$ is mad.

Definition

Let $\mathcal{A}$ be an infinite almost disjoint family on ω. We say that $\mathcal{A}$ is *completely separable* if for all $X \in \mathcal{I}^+(\mathcal{A})$, there exists some $A \in \mathcal{A}$ such that $A \subseteq X$.

Note that every completely separable almost disjoint family is mad.

Completely separable mad families on ω can be constructed under the following assumptions:

1. (Balcar-Simon, 1989) $\mathfrak{a} = \mathfrak{c}$ or $\mathfrak{b} = \mathfrak{d}$ or $\mathfrak{s} = \omega_1$.
2. (Simon, 1996) $\mathfrak{d} \leq \mathfrak{a}$.
3. (Shelah, 2011) $\mathfrak{s} < \mathfrak{a}$, or ($\mathfrak{a} \leq \mathfrak{s}$ + a certain PCF hypothesis).
   - For instance, $\mathfrak{c} < \aleph_\omega$ satisfies this PCF hypothesis.
4. (Mildenberger-Raghavan-Steprāns, 2014) $\mathfrak{s} \leq \mathfrak{a}$.

The existence of a completely separable mad family on ω in just ZFC remains open to date.

We give a sketch of the construction of a completely seperable mad family on ω.

Construction for $\mathfrak{s} \leq \mathfrak{a}$

An important component of the proof is the notion of a *block-splitting family*.

Definition

1. Let $\mathcal{P} = \{P_n : n < \omega\}$ be an interval partition. We say that a set $S \in [\omega]^\omega$ *block-splits* $\mathcal{P}$ if $\{n < \omega : P_n \subseteq S\}$ and $\{n < \omega : P_n \cap S = \emptyset\}$ are both infinite.
2. A *block-splitting family* is a family $\mathcal{S} \subseteq [\omega]^\omega$ such that every interval partition is block-split by some $S \in \mathcal{S}$.

Lemma (Kamburelis-Weglorz, 1996)

$\max\{\mathfrak{b}, \mathfrak{s}\} = \min\{|\mathcal{S}| : \mathcal{S}$ *is a block-splitting family*$\}$.

Lemma

Let $\{S_\alpha : \alpha < \max\{\mathfrak{b}, \mathfrak{s}\}\}$ be a block-splitting family. Let $\mathcal{A}$ be an almost disjoint family on ω, and let $X \in \mathcal{I}^+(\mathcal{A})$. There exists some α such that $X \cap S_\alpha \in \mathcal{I}^+(\mathcal{A})$ and $X \setminus S_\alpha \in \mathcal{I}^+(\mathcal{A})$.

Proof.

If there is some $Y \subseteq X$ that is almost disjoint from every element of $\mathcal{A}$, then let S_α be any set such that $Y \cap S_\alpha$ and $Y \setminus S_\alpha$ are infinite. Then $Y \cap S_\alpha, Y \setminus S_\alpha \in \mathcal{I}^+(\mathcal{A})$.

Otherwise, there is an infinite $\{A_i : n < \omega\} \subseteq \mathcal{A}$ such that $X \cap A_i$ is infinite for all i. Let $\mathcal{P} = \{P_n : n < \omega\}$ be an interval partition such that $P_n \cap (X \cap A_i) \neq \emptyset$ for all $i \leq n$. Let S_α block-splits $\mathcal{P}$.

1. $(X \cap S_\alpha) \cap A_i$ is infinite for all i.
2. $(X \setminus S_\alpha) \cap A_i$ is infinite for all i.

Thus, they're both in $\mathcal{I}^{++}(\mathcal{A}) \subseteq \mathcal{I}^+(\mathcal{A})$. $\square$

Assume that $\mathfrak{s} \leq \mathfrak{a}$. Note that $\mathfrak{b} \leq \mathfrak{a}$ holds in ZFC, so we have that $\max\{\mathfrak{b}, \mathfrak{s}\} \leq \mathfrak{a}$. Given $X \in [\omega]^\omega$, let $X^0 := X$ and $X^1 := \omega \setminus X$. Note that:

$$X \cap Y^0 = X \cap Y, \; X \cap Y^1 = X \setminus Y.$$

Let $\mathcal{S} = \{S_\alpha : \alpha < \max\{\mathfrak{b}, \mathfrak{s}\}\}$ be a block-splitting family. For a fixed almost disjoint family $\mathcal{A}$ and $X \in \mathcal{I}^+(\mathcal{A})$, let $\tau_X^{\mathcal{A}} \in 2^\alpha$ be the unique function where:

1. α is the least ordinal such that $X \cap S_\alpha \in \mathcal{I}^+(\mathcal{A})$ and $X \setminus S_\alpha \in \mathcal{I}^+(\mathcal{A})$.
2. If $\beta < \alpha$, then $X \cap S_\beta^{\tau_X^{\mathcal{A}}(\beta)} \in \mathcal{I}^+(\mathcal{A})$ (so $X \cap S_\beta^{1-\tau_X^{\mathcal{A}}(\beta)} \in \mathcal{I}(\mathcal{A})$).

Observe that if $Y \subseteq X$ are in $\mathcal{I}^+(\mathcal{A})$, then $\tau_X^{\mathcal{A}} \sqsubseteq \tau_Y^{\mathcal{A}}$.

Let $\{X_\alpha : \alpha < \mathfrak{c}\}$ cofinally enumerate $[\omega]^\omega$. We recursively construct our mad family $\mathcal{A} = \{A_\alpha : \alpha < \mathfrak{c}\}$ and $\{\sigma_\alpha : \alpha < \mathfrak{c}\} \subseteq 2^{<\max\{\mathfrak{b},\mathfrak{s}\}}$ such that (where $\mathcal{A}_\alpha := \{A_\beta : \beta < \alpha\}$):

1. If $\beta < \alpha$, then $\sigma_\alpha \not\subseteq \sigma_\beta$.
2. If $X_\alpha \in \mathcal{I}^+(\mathcal{A}_\alpha)$, then $A_\alpha \subseteq X_\alpha$.
3. If $\xi \in \mathrm{dom}(\sigma_\alpha)$, then $A_\alpha \subseteq^* S_\xi^{\sigma_\alpha(\xi)}$.

It's clear that if we are able to complete this construction, then $\mathcal{A}$ is completely separable.

Assume that $\mathcal{A}_\alpha = \{A_\beta : \beta < \alpha\}$ has been defined. We define $\{X_s : s \in 2^{<\omega}\} \subseteq \mathcal{I}^+(\mathcal{A}_\alpha)$ as follows:

1. $X_\emptyset := X_\alpha$ if $X_\alpha \in \mathcal{I}^+(\mathcal{A}_\alpha)$. Otherwise, $X_\emptyset := \omega$.
2. Let $\eta_s := \tau_{X_s}^{\mathcal{A}_\alpha}$ and $\alpha_s := \mathrm{dom}(\eta_s)$.
3. $X_{s^\frown 0} := X_s \cap S_{\alpha_s}$, $X_{s^\frown 1} := X_s \setminus S_{\alpha_s}$.

Note that if $t \sqsubseteq s$, then $X_s \subseteq X_t$, so $\eta_t \sqsubseteq \eta_s$.

For each $f \in 2^\omega$, let $\eta_f := \bigcup_{n<\omega} \eta_{f\restriction n}$. Note that $\eta_f \in 2^{<\max\{\mathfrak{b},\mathfrak{s}\}}$, as $\mathrm{cf}(\max\{\mathfrak{b},\mathfrak{s}\}) > \omega$. Since $\alpha < \mathfrak{c}$, there exists some $f \in 2^\omega$ such that $\eta_f \not\sqsubseteq \sigma_\beta$ for all β. We let $\sigma_\alpha := \eta_f$.

Observe that $(X_{f\restriction n})_{n<\omega}$ forms a $\subseteq$-decreasing sequence in $\mathcal{I}^+(\mathcal{A}_\alpha)$. Let $Y \in \mathcal{I}^+(\mathcal{A}_\alpha)$ be such that $Y \subseteq^* X_{f\restriction n}$ for all n.

Claim

There exists an almost disjoint family $\mathcal{C} \subseteq \mathcal{A}_\alpha$ such that:

1. $|\mathcal{C}| < \max\{\mathfrak{b}, \mathfrak{s}\}$.
2. *If $B \subseteq Y$ is almost disjoint from every element of $\mathcal{C}$, then B is almost disjoint from A_β for all $\beta < \alpha$.*

Now let:

$$\mathcal{C}' := \{A \cap Y : A \in \mathcal{C}\}.$$

Then $\mathcal{C}'$ is an almost disjoint family of subsets of Y.

Since $|\mathcal{C}'| < \max\{\mathfrak{b}, \mathfrak{s}\} \leq \mathfrak{a}$, there exists some $A_\alpha \subseteq Y$ that is almost disjoint from every element of $\mathcal{C}$. Then this A_α works.

Mad families of subspaces

We first extend our previously introduced definitions to almost disjoint families of subspaces.

Definition

Let $Y \subseteq E$.

1. Y is *big* if there exists some (infinite-dimensional) subspace $V \subseteq Y$.
2. Y is *small* if it is not big.
3. Y is *very small* if for all small $Z \subseteq E$, $Y \cup Z$ is small.

Note that if $\mathbb{F} \neq \mathbb{F}_2$, then finite union of small sets need not be small. In other words, small sets need not be very small.

Example

Fix some $\mu \in \mathbb{F} \setminus \{0, 1\}$. Let:

$$Y := \{x \in E : x = e_n + y \text{ for some } e_n < y\}.$$

For any subspace U and $x \in U$, $x = \lambda e_n + y$ for some $e_n < y$ and $0 \neq \lambda$. Then $\frac{1}{\lambda}x \in U \cap Y$, and $\frac{\mu}{\lambda}x \in U \cap Y^c$. Thus, Y and Y^c are small, but $E = Y \cup Y^c$ isn't.

Let $\mathcal{A}$ be an almost disjoint family of subspaces (of E). We let:

$$\mathcal{I}(\mathcal{A}) := \left\{ Y \subseteq E : Y \setminus \bigcup \mathcal{F} \text{ is small for some finite } \mathcal{F} \subseteq \mathcal{A} \right\}.$$

We also let:

$$\mathcal{I}^{+}(\mathcal{A}) := \mathcal{P}(E) \setminus \mathcal{I}(\mathcal{A}),$$
$$\mathcal{I}^{++}(\mathcal{A}) := \{ Y \subseteq E : Y \cap V \text{ is big for infinitely many } V \in \mathcal{A} \}.$$

Fact (Issue 1)

$\mathcal{I}(\mathcal{A})$ is "almost" an ideal.

Definition

Let $\mathcal{A}$ be an almost disjoint family of subspaces. We say that $\mathcal{A}$ is *completely separable* if for all subspaces $U \in \mathcal{I}^+(\mathcal{A})$, there exists some $V \in \mathcal{A}$ such that $V \subseteq U$.

Once again, every completely separable almost disjoint family of subspaces is maximal.

Remark. The requirement for U to be a subspace cannot be removed.

Completely separable mad families of subspaces

Theorem (Y., 2025)

There is a completely separable mad family of block subspaces.

We shall attempt to repeat the construction assuming $\mathfrak{s} \leq \mathfrak{a}$, then make the necessary adjustments along the way.

Let $\{S_\alpha : \alpha < \max\{\mathfrak{b}, \mathfrak{s}\}\}$ be a block-splitting family. We let:

$$Z_\alpha := \langle e_n : n \in S_\alpha \rangle .$$

Lemma (Issue 2)

Let $\mathcal{A}$ be an almost disjoint family of subspace, and let $\langle B \rangle \in \mathcal{I}^+(\mathcal{A})$ be a block subspace for some $B \in E^{[\infty]}$.

1. *For all α, $\langle B \rangle \cap Z_\alpha \in \mathcal{I}^+(\mathcal{A})$ or $\langle B \rangle \setminus Z_\alpha \in \mathcal{I}^+(\mathcal{A})$.*
2. *If $\langle B \rangle \cap Z_\alpha \in \mathcal{I}^+(\mathcal{A})$, then there exists some block subspace $\langle C \rangle \in \mathcal{I}^+(\mathcal{A})$ such that $\langle C \rangle \subseteq \langle B \rangle \cap Z_\alpha$.*
3. *If $\langle B \rangle \setminus Z_\alpha \in \mathcal{I}^+(\mathcal{A})$, then there exists some block subspace $\langle C \rangle \in \mathcal{I}^+(\mathcal{A})$ such that $\langle C \rangle \subseteq \langle B \rangle \setminus Z_\alpha$.*
4. *There exists some α such that $\langle B \rangle \cap Z_\alpha \in \mathcal{I}^+(\mathcal{A})$ and $\langle B \rangle \setminus Z_\alpha \in \mathcal{I}^+(\mathcal{A})$.*

One would begin by assuming that $\max\{\mathfrak{b}, \mathfrak{s}\} \leq \mathfrak{a}_{\mathrm{vec},\mathbb{F}}$ holds, but it turns out that this is unnecessary.

Lemma (Brendle-García Ávila, 2017 + Smythe, 2019)

$\mathrm{non}(\mathcal{M}) \leq \mathfrak{a}_{\mathrm{vec},\mathbb{F}}$.

Since $\max\{\mathfrak{b}, \mathfrak{s}\} \leq \mathrm{non}(\mathcal{M})$ holds in ZFC, we have that $\max\{\mathfrak{b}, \mathfrak{s}\} \leq \mathfrak{a}_{\mathrm{vec},\mathbb{F}}$.

Let $\{Z_\alpha : \alpha < \max\{\mathfrak{b}, \mathfrak{s}\}\}$ be the family of block subspaces previously defined. For a fixed almost disjoint family of subspaces $\mathcal{A}$ and block subspace $\langle B \rangle \in \mathcal{I}^+(\mathcal{A})$, let $\tau_B^{\mathcal{A}} \in 2^\alpha$ be the unique function where:

1. α is the least ordinal such that $\langle B \rangle \cap Z_\alpha \in \mathcal{I}^+(\mathcal{A})$ and $\langle B \rangle \setminus Z_\alpha \in \mathcal{I}^+(\mathcal{A})$.
2. If $\beta < \alpha$, then $\langle B \rangle \cap Z_\beta^{\tau_B^{\mathcal{A}}(\beta)} \in \mathcal{I}^+(\mathcal{A})$ (so $\langle B \rangle \cap Z_\beta^{1-\tau_B^{\mathcal{A}}(\beta)} \in \mathcal{I}(\mathcal{A})$).

Observe that if $\langle C \rangle \subseteq \langle B \rangle$ are in $\mathcal{I}^+(\mathcal{A})$, then $\tau_B^{\mathcal{A}} \sqsubseteq \tau_C^{\mathcal{A}}$.

Let $\{U_\alpha : \alpha < \mathfrak{c}\}$ cofinally enumerate all subspaces of E. We recursively construct our mad family of block subspaces $\mathcal{A} = \{A_\alpha : \alpha < \mathfrak{c}\}$ and $\{\sigma_\alpha : \alpha < \mathfrak{c}\} \subseteq 2^{<\max\{\mathfrak{b},\mathfrak{s}\}}$ such that (where $\mathcal{A}_\alpha := \{A_\beta : \beta < \alpha\}$):

1. If $\beta < \alpha$, then $\sigma_\alpha \not\sqsubseteq \sigma_\beta$.
2. If $U_\alpha \in \mathcal{I}^+(\mathcal{A}_\alpha)$, then $\langle A_\alpha \rangle \subseteq U_\alpha$.
3. If $\xi \in \mathrm{dom}(\sigma_\alpha)$, then $\langle A_\alpha \rangle \setminus Z_\xi^{\sigma_\alpha(\xi)}$ is small.

It's clear that if we are able to complete this construction, then $\mathcal{A}$ is completely separable.

Assume that $\mathcal{A}_\alpha = \{A_\beta : \beta < \alpha\}$ has been defined. We define block sequences $\{C_s : s \in 2^{<\omega}\}$ as follows:

1. (**Issue 3**) If $U_\alpha \in \mathcal{I}^+(\mathcal{A})$, let $\langle C_\emptyset \rangle \in \mathcal{I}^+(\mathcal{A}_\alpha)$ be a block subspace so that $\langle C_\emptyset \rangle \subseteq U_\alpha$ (otherwise, let $\langle C_\emptyset \rangle = E$).
2. Let $\eta_s := \tau^{\mathcal{A}_\alpha}_{C_s}$ and $\alpha_s := \mathrm{dom}(\eta_s)$.
3. $\langle C_{s^\frown 0} \rangle, \langle C_{s^\frown 1} \rangle \in \mathcal{I}^+(\mathcal{A})$ such that $\langle C_{s^\frown 0} \rangle \subseteq \langle C_s \rangle \cap Z_{\alpha_s}$ and $\langle C_{s^\frown 1} \rangle \subseteq \langle C_s \rangle \setminus Z_{\alpha_s}$.

Note that if $t \sqsubseteq s$, then $C_s \leq C_t$ (i.e. $\langle C_s \rangle \subseteq \langle C_t \rangle$), so $\eta_t \sqsubseteq \eta_s$.

For each $f \in 2^\omega$, let $\eta_f := \bigcup_{n<\omega} \eta_{f\restriction n}$. Note that $\eta_f \in 2^{<\max\{\mathfrak{b},\mathfrak{s}\}}$. Let $f \in 2^\omega$ be such that $\eta_f \not\sqsubseteq \sigma_\beta$ for all β. We let $\sigma_\alpha := \eta_f$.

Claim (**Issue 4**)

There exists some block sequence $\langle D \rangle \in \mathcal{I}^+(\mathcal{A}_\alpha)$ such that $D \leq^* C_{f\restriction n}$ for all n (i.e. $\langle D \rangle \setminus \langle C_{f\restriction n} \rangle$ is small).

Claim

There exists an almost disjoint family $\mathcal{C} \subseteq \mathcal{A}_\alpha$ such that:

1. $|\mathcal{C}| < \max\{\mathfrak{b}, \mathfrak{s}\}$.
2. *If $B \leq D$ is almost disjoint from every element of $\mathcal{C}$, then B is almost disjoint from A_β for all $\beta < \alpha$.*

Now let:

$$\mathcal{C}' := \{\langle A\rangle \cap \langle D\rangle : \langle A\rangle \in \mathcal{A}\}.$$

Lemma

Finite intersection of block subspaces is a block subspace. Therefore, $\mathcal{C}'$ is an almost disjoint family of block subspaces of $\langle D\rangle$.

Since $|\mathcal{C}'| < \max\{\mathfrak{b}, \mathfrak{s}\} \leq \mathfrak{a}_{\mathrm{vec},\mathbb{F}}$, there exists some $A_\alpha \leq D$ that is almost disjoint from every element of $\mathcal{C}$. Then this A_α works.

Resolving the issues

Summary of issues:

1. Show that $\mathcal{I}(\mathcal{A})$ is "almost" an ideal.
2. The main lemma about $\{Z_\alpha : \alpha < \max\{\mathfrak{b}, \mathfrak{s}\}\}$.
3. If $U \in \mathcal{I}^+(\mathcal{A})$, then there exists some block subspace $\langle B \rangle \in \mathcal{I}^+(\mathcal{A})$ such that $\langle B \rangle \subseteq U$.
4. If $C_0 \geq C_1 \geq \cdots$ are in $\mathcal{I}^+(\mathcal{A})$, then there exists some $\langle D \rangle \in \mathcal{I}^+(\mathcal{A})$ such that $D \leq^* C_n$ for all n.

Issue 1: $\mathcal{I}(\mathcal{A})$ is "almost" an ideal.

We first prove a basic lemma. Recall that a set $Y \subseteq E$ is *very small* if for all small $Z \subseteq E$, $Y \cup Z$ is small.

Lemma

Let U, V be infinite-dimensional subspaces.

1. *If $U \cap V$ is small, then $U \cap V$ is very small.*
2. *If $U \setminus V$ is small, then $U \setminus V$ is very small.*

Consequently, we cannot have both $U \cap V$ and $U \setminus V$ small.

Proof.

1. Let $Y \subseteq E$, and suppose that $W \subseteq (U \cap V) \cup Y$ for some subspace W. Since $U \cap V$ is a finite-dimensional subspace, let $W' \subseteq W$ be a subspace such that $W' \cap (U \cap V) = \emptyset$. Then $W' \subseteq Y$, so Y is big.
2. Let $Y \subseteq E$, and suppose that $W \subseteq (U \setminus V) \cup Y$ for some subspace W.
   - 2.1 If $W \cap U$ is small, then let $W' \subseteq W$ be a subspace such that $W' \cap (W \cap U) = \emptyset$. Then $W' \subseteq Y$, so Y is big.
   - 2.2 If $W \cap V$ is big, then $W \cap V \subseteq Y$, so Y is big.
   - 2.3 If $W \cap U$ is big but $W \cap V$ is small, then let $W' \subseteq W \cap U$ be such that $W' \cap V = \emptyset$. Then $W \subseteq U \setminus V$, a contradiction.

$\square$

Lemma

$\mathcal{I}(\mathcal{A})$ is "almost" an ideal in the following sense:

1. *$E \notin \mathcal{I}(\mathcal{A})$.*
2. *If $U \in \mathcal{I}(\mathcal{A})$, and $W \subseteq U$, then $W \in \mathcal{I}(\mathcal{A})$.*
3. *If $U, W \in \mathcal{I}(\mathcal{A})$ are two subspaces, then $U \cup W \in \mathcal{I}(\mathcal{A})$.*

Note that (2) is obvious by definition.

Proof.

1. Suppose that $E \in \mathcal{I}(\mathcal{A})$, so $E \setminus \bigcup_{i<n} V_i$ is small for some $V_0, \ldots, V_{n-1} \in \mathcal{A}$. Let $V_n \in \mathcal{A} \setminus \{V_0, \ldots, V_{n-1}\}$. Then:

$$V_n = \left(V_n \setminus \bigcup_{i<n} V_i \right) \cup \bigcup_{i<n} V_n \cap V_i,$$

which is small, a contradiction.

Proof (Cont.).

3. Let $V_0, \ldots, V_{n-1} \in \mathcal{A}$ be such that $U \setminus \bigcup_{i<n} V_i$ and $W \setminus \bigcup_{i<n} V_i$ are small. We shall show that $(U \cup W) \setminus \bigcup_{i<n} V_i$. Otherwise, suppose that $W' \subseteq (U \cup W) \setminus \bigcup_{i<n} V_i$. Then:

$$W' \cap U \subseteq U \setminus \bigcup_{i<n} V_i,$$

$$W' \setminus U \subseteq W \setminus \bigcup_{i<n} V_i,$$

are both small, a contradiction.

$\square$

Issue 2: Main lemma

We recall the lemma. Let $\{S_\alpha : \alpha < \max\{\mathfrak{b}, \mathfrak{s}\}\}$ be a block-splitting family. We let $Z_\alpha := \langle e_n : n \in S_\alpha \rangle$.

Lemma

Let $\mathcal{A}$ be an almost disjoint family of subspace, and let $\langle B \rangle \in \mathcal{I}^+(\mathcal{A})$ be a block subspace for some $B \in E^{[\infty]}$.

1. *For all α, $\langle B \rangle \cap Z_\alpha \in \mathcal{I}^+(\mathcal{A})$ or $\langle B \rangle \setminus Z_\alpha \in \mathcal{I}^+(\mathcal{A})$.*
2. *If $\langle B \rangle \cap Z_\alpha \in \mathcal{I}^+(\mathcal{A})$, then there exists some block subspace $\langle C \rangle \in \mathcal{I}^+(\mathcal{A})$ such that $\langle C \rangle \subseteq \langle B \rangle \cap Z_\alpha$.*
3. *If $\langle B \rangle \setminus Z_\alpha \in \mathcal{I}^+(\mathcal{A})$, then there exists some block subspace $\langle C \rangle \in \mathcal{I}^+(\mathcal{A})$ such that $\langle C \rangle \subseteq \langle B \rangle \cap Z_\alpha$.*
4. *There exists some α such that $\langle B \rangle \cap Z_\alpha \in \mathcal{I}^+(\mathcal{A})$ and $\langle B \rangle \setminus Z_\alpha \in \mathcal{I}^+(\mathcal{A})$.*

Lemma

1. *For all* α, $\langle B\rangle \cap Z_\alpha \in \mathcal{I}^+(\mathcal{A})$ *or* $\langle B\rangle \setminus Z_\alpha \in \mathcal{I}^+(\mathcal{A})$.

Proof of (1).

Suppose for a contradiction that there are $V_0, \ldots, V_{n-1} \in \mathcal{A}$ such that $(\langle B\rangle \cap Z_\alpha) \setminus \bigcup_{i<n} V_i$ and $(\langle B\rangle \setminus Z_\alpha) \setminus \bigcup_{i<n} V_i$ are small. Since $\langle B\rangle \in \mathcal{I}^+(\mathcal{A})$, $U \subseteq \langle B\rangle \setminus \bigcup_{i<n} V_i$ for some subspace U. Then:

$$U \cap Z_\alpha \subseteq (\langle B\rangle \cap Z_\alpha) \setminus \bigcup_{i<n} V_i,$$

$$U \setminus Z_\alpha \subseteq (\langle B\rangle \setminus Z_\alpha) \setminus \bigcup_{i<n} V_i,$$

are both small, a contradiction. $\square$

Lemma

4. *There exists some α such that $\langle B \rangle \cap Z_\alpha \in \mathcal{I}^+(\mathcal{A})$ and $\langle B \rangle \setminus Z_\alpha \in \mathcal{I}^+(\mathcal{A})$.*

Proof.

Suppose that there is some $C \leq B$ that is almost disjoint from every element of $\mathcal{A}$. Write $C = (x_n)_{n<\omega}$. Let $\{P_n : n < \omega\}$ be an interval partition such that $\mathrm{supp}(x_n) \subseteq P_n$ for all n. This means that:

1. For infinitely many n, $\mathrm{supp}(x_n) \subseteq S_\alpha$, so $x_n \in Z_\alpha = \langle (e_n)_{n \in S_\alpha} \rangle$. Therefore, $C \cap Z_\alpha$ is an infinite-dimensional subspace almost disjoint from every element of $\mathcal{A}$, so $C \cap Z_\alpha \in \mathcal{I}^+(\mathcal{A})$.
2. For infinitely many n, $\mathrm{supp}(x_n) \subseteq S_\alpha^c$. Thus:

$$x_n \in \langle (e_n)_{n \notin S_\alpha} \rangle \subseteq Z_\alpha^c.$$

   Therefore, $C \setminus Z_\alpha$ contains $C \cap \langle (e_n)_{n \notin S_\alpha} \rangle$, an infinite-dimensional subspace almost disjoint from every element of $\mathcal{A}$, so $C \setminus Z_\alpha \in \mathcal{I}^+(\mathcal{A})$.

Proof (Cont.).

Suppose that for all $C \leq B$, C is not almost disjoint from some element of $\mathcal{A}$. Let $\{V_i : i < \omega\} \subseteq \mathcal{A}$ be such that $\langle B \rangle \cap V_i$ is big for all n.

Fix an infinite-to-one surjection $\Gamma : \omega \to \omega$. We define a block sequence $(x_n)_{n<\omega}$ as follows: Let $x_0 \in \langle B \rangle \cap V_{\Gamma(0)}$. If x_n has been defined, then let $x_{n+1} \in \langle B \rangle \cap V_{\Gamma(n+1)}$ such that $x_n < x_{n+1}$.

Proof (Cont.).

We define an interval partition $\mathcal{P} = \{P_m : m < \omega\}$ as follows:

1. Let $P_0 := \{0, \ldots, \max(\mathrm{supp}(x_{n_0}))\}$, where n_0 is the least integer so that $\Gamma(n_0) = 0$.
2. If P_m has been defined, then let P_{m+1} be an interval such that for all $i \leq m+1$, there exists some $n > \max(P_m)$ such that $\Gamma(n) = i$ and $\mathrm{supp}(x_n) \subseteq P_{m+1}$.

Now suppose that S_α splits $\mathcal{P}$. For each i, let:

$$T_i^0 := \{n < \omega : \Gamma(n) = i \wedge \mathrm{supp}(x_n) \subseteq S_\alpha\},$$
$$T_i^1 := \{n < \omega : \Gamma(n) = i \wedge \mathrm{supp}(x_n) \cap S_\alpha = \emptyset\}.$$

By our choice of $\mathcal{P}$, T_i^0, T_i^1 are infinite for all i. This implies that $V_i \cap (B \cap Z_\alpha)$ and $V_i \cap (B \setminus Z_\alpha)$ are big for all i, so $B \cap Z_\alpha, B \setminus Z_\alpha \in \mathcal{I}^+(\mathcal{A})$.

Thank you for listening!

1. Two (infinite-dimensional) subspaces are almost disjoint if their intersection is finite-dimensional.
2. A mad family of subspaces $\mathcal{A}$ is *completely separable* if for all subspaces $U \in \mathcal{I}^+(\mathcal{A})$, there exists some $V \in \mathcal{A}$ such that $V \subseteq U$.
3. $\max\{\mathfrak{b}, \mathfrak{s}\} \leq \mathfrak{a}_{\mathrm{vec},\mathbb{F}}$ holds in ZFC. Consequently, following the construction of a completely separable mad family on ω assuming $\mathfrak{s} \leq \mathfrak{a}$, we can show that there is a completely separable mad family of subspaces in ZFC.
4. Several issues needed to be and were addressed when replicating the construction.
   - 4.1 While $\mathcal{I}(\mathcal{A})$ need not be an ideal, it is "almost" an ideal.
   - 4.2 If $Z_\alpha := \{S_\alpha : \alpha < \max\{\mathfrak{b}, \mathfrak{s}\}\}$, then Z_α splits $\langle B \rangle \in \mathcal{I}^+(\mathcal{A})$ into two $\mathcal{I}(\mathcal{A})$-positive sets for some α.